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# Fairness Auditing of an Income-Classification Model: SHAP-Based Analysis of Proxy Features

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## Abstract

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## 1 Introduction

Machine-learning models can produce unequal outcomes across demographic groups even when their average predictive performance is high [Mehrabi et al., 2021]. A common response is to remove a sensitive attribute, such as *sex*, from the model inputs. This intervention is simple and often presented as a way to prevent discrimination. However, other features may still contain information associated with the protected attribute (i.e., *sex*) and allow the model to reproduce group disparities [Datta et al., 2017]. Removing *sex* therefore does not by itself establish that a model is fair.

This paper studies that problem in income classification, where *sex* is treated as a sensitive feature because it marks a socially salient group for which historical and structural inequalities can affect access to education, occupations, wages, and other opportunities represented in the data [Mehrabi et al., 2021, Barocas et al., 2019]. The concern is not that *sex* is inherently inappropriate for prediction or that it should never be observed; rather, using it, or information that acts as its proxy, may allow a model to reproduce or amplify unequal treatment in decisions related to economic opportunity. We use the UCI Adult Income dataset, which contains demographic, educational, occupational, and work-related features and a binary target indicating whether annual income exceeds \$50,000 [Dua and Graff, 2019]. The dataset provides a useful setting for a controlled audit because it contains an explicit *sex* attribute alongside several features that may encode social and economic structure.

In this study, we ask whether excluding *sex* changes the fairness of an income-classification model and whether remaining features can act as proxies for it. To answer this question, we (1) compare models trained with and without *sex*, (2) investigate whether selected remaining features contribute to group-related disparities, and (3) compare feature removal with fairness-aware mitigation. Predictive performance and group fairness are evaluated together so that an apparent fairness improvement is not interpreted without considering its effect on utility. An overview of our method is provided in Figure 1. This paper makes three contributions:

- We provide a reproducible comparison of income-classification models trained with and without the explicit sensitive attribute *sex*.

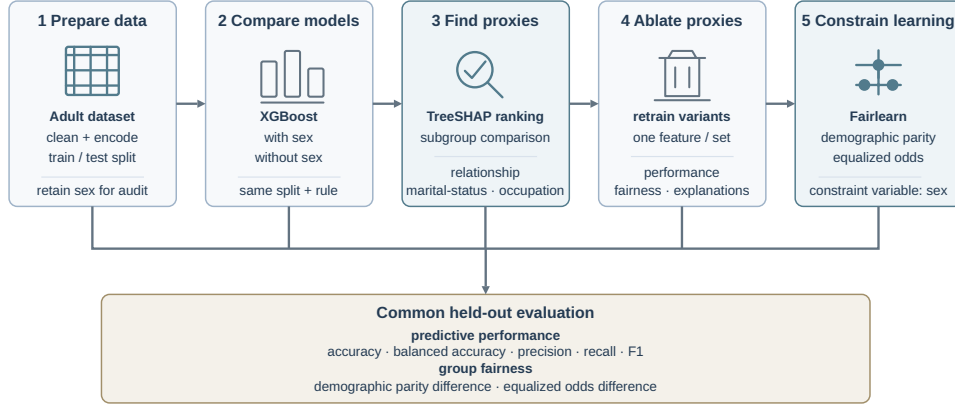


Figure 1: Overview of our proposed method. The audit compares explicit-feature removal, proxy ablation, and fairness-constrained learning under a common held-out evaluation.

- We combine model explanations, subgroup comparisons, and feature ablation to investigate whether specific remaining features behave as potential proxies.
- We compare proxy removal with fairness-constrained learning using common predictive and group-fairness metrics, making the associated performance–fairness trade-offs explicit.

The remainder of this paper is structured as follows. Section 2 discusses related work; section 3 describes the proposed method, dataset, and evaluation metrics; section 4 presents the experimental results; section 5 discusses the limitations and future work; finally, section 6 concludes.

## 2 Related Work

**Fairness through unawareness, proxy discrimination, and causal fairness.** One influential response to discriminatory prediction is to omit protected attributes from the model. However, Dwork et al. [2012] argue that fairness cannot be reduced to ignoring sensitive attributes: their *Fairness Through Awareness* framework instead requires similar individuals, under a task-specific similarity metric, to receive similar treatment. This distinction is directly relevant to our comparison between models with and without sex: removing the column is an unawareness baseline, not evidence that the remaining representation is fair. The proxy discrimination literature makes the same concern operational by studying how protected-group information can be inferred from seemingly permissible features and how such relationships can be audited [Datta et al., 2017]. By contrast, Kusner et al. [2017] define counterfactual fairness through a causal model, requiring a prediction to remain unchanged in counterfactual worlds in which an individual’s protected attribute differs. Our SHAP rankings, subgroup comparisons, and ablations are deliberately more modest: they can identify predictive association and model dependence on candidate proxies, but they cannot determine whether a feature is a causal mediator, a legitimate correlate, or an impermissible proxy.

**Group fairness criteria and the problem of incompatible objectives.** A second body of work formalizes how disparities should be measured once a model has been trained. Hardt et al. [2016] introduce equality of opportunity and equalized odds, connecting fairness to conditional error rates and providing a post-processing procedure that uses group membership to alter predictions. These ideas motivate our reporting of true- and false-positive rate disparities in addition to selection-rate disparities. Yet no single criterion resolves the broader normative question. Chouldechova [2017] show that calibration and error-rate parity can conflict when outcome prevalence differs between groups, while Kleinberg et al. [2017] prove closely related impossibility results for risk scores. Critical reviews likewise emphasize that criteria such as statistical parity encode different values and can lead to different conclusions about the same system [Corbett-Davies and Goel, 2017, Barocas et al., 2019]. Accordingly, our audit reports demographic-parity difference and equalized-odds difference as separate descriptive outcomes alongside accuracy, balanced accuracy, precision, recall,

and F1. The Adult experiment therefore tests whether an intervention changes particular measured disparities; it does not treat a small value for one metric as a complete definition of fairness.

**Explainability and feature-level auditing for proxy detection.** The third relevant line of work uses explanations to inspect how models reach their predictions. Lundberg and Lee [2017] introduce SHAP, an additive feature-attribution framework grounded in Shapley values whose local accuracy and consistency properties make it attractive for comparing contributions across individual predictions. We use TreeSHAP in this spirit to rank features in the sex-excluded XGBoost model and to prioritize variables such as `relationship`, `marital-status`, `occupation`, and `hours-per-week` for further investigation. SHAP, however, is an explanatory device rather than a fairness test: a large attribution indicates model reliance, not discriminatory intent or causal influence. This caveat is especially important for Adult because its features are dependent. Aas et al. [2021] show that common SHAP approximations can be misleading when feature independence assumptions fail and propose dependence-aware approximations and grouped explanations. Our use of explanations is therefore paired with subgroup comparisons and feature ablation. The combination asks a narrower and testable question than explanation alone: whether removing a candidate feature changes both model behavior and observed group disparities on the same held-out population.

**Fairness-aware mitigation and constrained learning.** Feature deletion is only one possible intervention, and it may remove useful predictive information without directly controlling the disparity of interest. Zafar et al. [2017] incorporate fairness constraints into the geometry of a classifier’s decision boundary, making the fairness–accuracy trade-off an explicit part of training. Madras et al. [2018] instead learn representations that make sensitive attributes difficult to predict using an adversary, relating the training objective to demographic parity, equal opportunity, and equalized odds. The reductions approach of Agarwal et al. [2018] is particularly close to our implementation: it converts constrained binary classification into a sequence of cost-sensitive classification problems and can optimize error subject to demographic-parity or equalized-odds constraints. This provides the conceptual basis for Fairlearn’s `ExponentiatedGradient` comparison in our audit. Our design places these approaches beside, rather than above, sex removal and proxy ablation: all variants use the same fixed split and held-out metrics, allowing us to distinguish fairness obtained by deleting features from fairness obtained by explicitly constraining predictions. The comparison also preserves `sex` for evaluation and mitigation constraints while excluding it from ordinary predictive features, so that the fairness intervention is measurable without treating the sensitive attribute as a standard model input.

Taken together, prior work establishes four points that motivate our study: omitting a sensitive feature does not rule out proxy discrimination; fairness criteria measure different and sometimes incompatible properties; explanations can reveal model reliance but do not by themselves establish causality or unfairness; and constrained learners offer a direct alternative to feature deletion. The gap addressed here is an integrated, reproducible audit that connects these pieces in one Adult-income experiment: it compares explicit sex removal with targeted proxy ablation, uses SHAP and subgroup diagnostics to justify which proxies are tested, and evaluates both feature interventions and Fairlearn reductions under common predictive and group-fairness metrics. This makes the practical claim more precise than either “remove the sensitive attribute” or “optimize a fairness metric”: the question is which intervention changes which observed disparities, at what performance and explainability cost, under the stated dataset and audit limitations.

### 3 Methodology

#### 3.1 Our proposed method

Figure 1 summarizes the proposed fairness audit as a five-stage workflow: prepare the UCI Adult Income dataset, compare XGBoost models with and without `sex`, identify and ablate candidate proxy features, and apply fairness constraints before evaluating every variant on the same held-out test set.

**Prepare the data.** We first clean and encode the dataset using a deterministic preprocessing pipeline. Preprocessing is fitted on the training data only and then applied to the held-out data to prevent leakage. The sensitive feature, `sex`, is retained outside the model feature matrix so that it remains

Table 1: Summary Statistics of the UCI Adult Income Dataset.

| Statistic                  | Value                         |
|----------------------------|-------------------------------|
| Total records              | 48,842                        |
| Training/test records      | 32,561 / 16,281               |
| Features                   | 14 (6 numeric, 8 categorical) |
| At or below \$50,000       | 37,155 (76.07%)               |
| Above \$50,000             | 11,687 (23.93%)               |
| Male/female records        | 32,650 / 16,192               |
| Male above-\$50K rate      | 30.38%                        |
| Female above-\$50K rate    | 10.93%                        |
| Missing categorical values | 6,465                         |

available for subgroup auditing even when it is excluded from model training. Further details about the dataset and preprocessing procedure are given in 3.2.

**Compare models with and without sex.** We train XGBoost models using the same data split, preprocessing, and decision rule. One baseline includes `sex`, while a paired model removes only that feature. This comparison isolates the effect of explicit feature removal while leaving the model family and experimental conditions unchanged. The sex-excluded model is therefore treated as an unawareness baseline, not as evidence that the remaining features are free of sensitive information.

**Identify candidate proxy features.** We analyze the sex-excluded model with TreeSHAP and subgroup comparisons [Lundberg and Lee, 2017]. Features are treated as candidate proxies when they both contribute to predictions and show relevant differences across the observed sex groups. This analysis focuses on features such as `relationship`, `marital-status`, `occupation`, and `hours-per-week`. These diagnostics identify model associations rather than establish causal or normative claims about any feature.

**Ablate proxies and retrain model variants.** We test these candidate proxies through controlled ablation. Individual features and selected feature sets are removed, and the XGBoost model is retrained under the same conditions. Comparing the resulting variants with the sex-excluded model shows whether proxy removal changes model performance, group disparities, or feature-level explanations.

**Apply fairness-constrained learning.** We compare feature removal with Fairlearn’s ExponentiatedGradient reduction [Agarwal et al., 2018]. The constrained models use `sex` as the constraint variable and impose either demographic-parity or equalized-odds constraints, rather than using it as an ordinary model feature. This provides a direct comparison between proxy-focused feature interventions and fairness-constrained learning.

**Common held-out evaluation.** Finally, all model variants are evaluated on the same held-out data using a common prediction rule. We compare predictive performance and group fairness for the baseline, sex-excluded, proxy-ablated, and constrained models. The specific evaluation measures and their definitions are described in 3.3.

### 3.2 Dataset

We use the UCI Adult Income dataset [Dua and Graff, 2019], preserving its official training and test partitions. The dataset contains demographic, educational, occupational, and work-related variables, with a binary target indicating whether annual income exceeds \$50,000. The sensitive attribute `sex` is retained for subgroup evaluation even when it is excluded from model features. The main descriptive statistics are summarized in Table 1. The table records the dataset size, the official partition sizes, the six numeric and eight categorical features, the imbalanced target distribution, the observed sex-group sizes, and the number of categorical values encoded as `Missing`. Thus, it gives the scale and composition of the data used throughout the audit without presenting any model outcome.

The target distribution and the observed group-specific rates are shown in Figure 2. The left panel shows that 11,687 of the 48,842 records (23.93%) are above \$50,000, so overall accuracy alone could

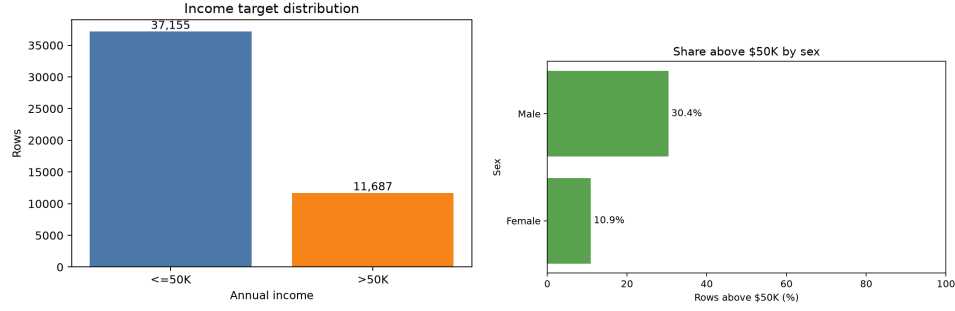


Figure 2: Income Distribution and Above-\$50K Rates Across Sex Groups. The panels summarize the cleaned dataset: the target distribution is shown on the left, and the observed proportion above \$50,000 across the two observed `sex` categories is shown on the right. These are descriptive dataset statistics, not model predictions or fairness results.

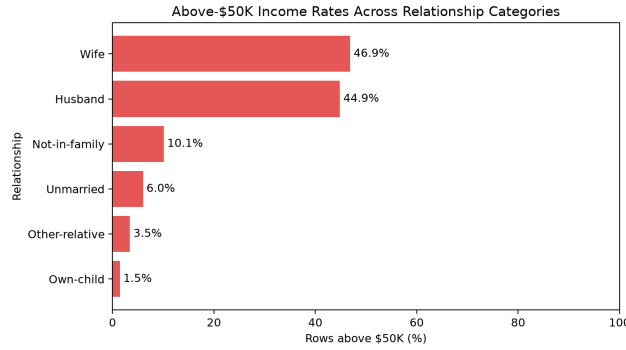


Figure 3: Above-\$50K Income Rates Across Relationship Categories. The plot reports observed target rates for each `relationship` category in the cleaned dataset. It is included as descriptive information and as a reference for a feature suspected of retaining group-related information; it does not by itself establish that `relationship` is a proxy for `sex` or that the observed differences are caused by the feature.

obscure performance on the less frequent class. The right panel shows that the above-\$50K rate is 30.38% for males and 10.93% for females. This is an observed association in the dataset, not a model prediction or a fairness result; it motivates retaining `sex` for evaluation even when it is removed from the model inputs.

The `relationship` feature is examined descriptively in Figure 3 because it represents family and household structure and is one of the remaining variables that could retain group-related information. The observed above-\$50K rates range from 1.46% for `Own-child` to 46.89% for `Wife`. These differences show that the feature has substantial structure relevant to the target, but they do not establish that it is a proxy for `sex`, identify a causal mechanism, or show that a trained model relies on it. We therefore use this plot as information and as a reference for the later SHAP, subgroup, and ablation analyses. Together, the explicit sensitive attribute, the observed target disparity, and the presence of potentially related family and work-related features make `Adult` suitable for a controlled proxy-feature audit, while the conclusions remain limited to the observed groups and dataset representation.

### 3.3 Evaluation metrics

We evaluate every model on the same held-out test split using predictive performance and group fairness metrics. Let  $Y \in \{0, 1\}$  denote the true income label,  $\hat{Y}$  the binary prediction, and  $A = \text{sex}$  the sensitive attribute. The observed groups are indexed by  $g \in \mathcal{G}$ , and all metrics use the same fixed decision rule.

**Predictive performance.** We report accuracy (Acc), balanced accuracy (BAcc), precision (Pr), recall (Rc), and F1. Accuracy is the fraction of all test records classified correctly. Because the above-\$50,000 class is less frequent, we also report balanced accuracy:

$$\text{BAcc} = \frac{\text{Sensitivity} + \text{Specificity}}{2} = \frac{1}{2} \left[ \frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} \right],$$

where sensitivity is the true-positive rate for the above-\$50,000 class and specificity is the true-negative rate for the at-or-below-\$50,000 class. BAcc gives equal weight to the two income classes and is therefore more informative than Acc when class frequencies are unequal. Precision is the fraction of positive predictions that are correct, Rc is the fraction of actual above-\$50,000 records identified, and F1 is their harmonic mean. Together, these measures distinguish overall correctness from the model’s ability to identify the minority income class and the reliability of its positive predictions.

**Group fairness.** We report demographic-parity difference (DPD) and equalized-odds difference (EOD) across the observed sex groups. As an auxiliary quantity for interpreting DPD and EOD, we also report the group-specific selection rate

$$\text{SR}_g = P(\hat{Y} = 1 \mid A = g),$$

which is the proportion of records in group  $g$  receiving a positive prediction. DPD compares positive prediction rates without conditioning on the true income label:

$$\text{DPD} = \max_{g \in \mathcal{G}} P(\hat{Y} = 1 \mid A = g) - \min_{g \in \mathcal{G}} P(\hat{Y} = 1 \mid A = g).$$

This metric is relevant to the audit because it measures whether the model assigns the positive income classification at different rates across groups, which is the disparity that may remain after removing sex from the features. It does not, however, account for differences in the observed target distribution. With the difference defined over probabilities, DPD has range  $[0, 1]$ : 0 indicates identical positive prediction rates, whereas larger values indicate greater disparity, with 1 being the maximum possible difference.

EOD measures disparities conditional on the true label:

$$\text{EOD} = \max \left\{ \max_{g \in \mathcal{G}} \text{TPR}_g - \min_{g \in \mathcal{G}} \text{TPR}_g, \max_{g \in \mathcal{G}} \text{FPR}_g - \min_{g \in \mathcal{G}} \text{FPR}_g \right\},$$

where  $\text{TPR}_g = P(\hat{Y} = 1 \mid Y = 1, A = g)$  is the group-specific true-positive rate and  $\text{FPR}_g = P(\hat{Y} = 1 \mid Y = 0, A = g)$  is the false-positive rate. EOD is suitable here because it checks whether groups with the same observed income label experience different opportunities for correct positive classification or different rates of erroneous positive classification. DPD and EOD therefore capture complementary notions: DPD measures disparity in selection, whereas EOD measures disparity in conditional error outcomes. Because both are differences of probabilities, EOD also has range  $[0, 1]$ . An EOD of 0 indicates identical TPR and FPR across the observed groups, whereas larger values indicate greater disparity in at least one of these rates, with 1 being the maximum possible difference. Thus, lower DPD and EOD values indicate less observed group disparity, not necessarily an unbiased or fair model. Zero is not a complete definition of fairness, particularly given the dataset’s binary sex representation and the limitations of any single fairness criterion.

**Proxy identification.** For proxy screening, we report the feature–sex association  $V_{\text{sex}}$ . For a categorical feature with contingency-table chi-square statistic  $\chi^2$ , sample size  $n$ , and table dimensions  $r \times c$ , its uncorrected form is

$$V_{\text{sex}} = \sqrt{\frac{\chi^2}{n \min(r - 1, c - 1)}}.$$

We use the bias-corrected Cramér’s  $V$  implementation for categorical features. For numeric features, we use the absolute Spearman correlation

$$V_{\text{sex}} = |\rho_{\text{Spearman}}(X, A)|,$$

Table 2: Predictive performance of XGBoost models trained with and without the `sex` feature on the fixed held-out test split ( $n = 16,281$ ). Difference is computed as without `sex` minus with `sex`.

| Metric | With <code>sex</code> | Without <code>sex</code> | Difference |
|--------|-----------------------|--------------------------|------------|
| Acc    | 0.8746                | 0.8748                   | +0.0001    |
| BAcc   | 0.7990                | 0.7996                   | +0.0005    |
| Pr     | 0.7786                | 0.7783                   | -0.0004    |
| Rc     | 0.6557                | 0.6570                   | +0.0013    |
| F1     | 0.7119                | 0.7125                   | +0.0006    |

where  $A$  is the binary indicator of the observed `sex` group. Both versions range from 0 to 1, with larger values indicating stronger univariate association. This association does not establish causation or determine whether the model relies on the feature.

To quantify model reliance, we use TreeSHAP [Lundberg and Lee, 2017]. For a model  $f$  and input  $x$ , the TreeSHAP decomposition is

$$f(x) = \phi_0 + \sum_{j=1}^p \phi_j(x),$$

where  $\phi_0$  is the baseline output and

$$\phi_j(x) = \sum_{S \subseteq \{1, \dots, p\} \setminus \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} [f_{S \cup \{j\}}(x) - f_S(x)]$$

is the Shapley contribution of feature  $j$ . We summarize model reliance using the mean absolute contribution,  $|\overline{\text{SHAP}}_j| = n^{-1} \sum_i |\phi_j(x_i)|$ , aggregating one-hot encoded contributions to the original feature. A candidate proxy is therefore a feature that is both used by the model and associated with the observed `sex` groups; neither diagnostic alone is sufficient.

## 4 Results

### 4.1 XGBoost With and Without the Sensitive Feature

We first compared two XGBoost models on the same held-out test split ( $n = 16,281$ ): a model using all features, including `sex`, and a paired model excluding only `sex`. The sensitive attribute was retained for evaluation in both cases. Table 2 reports the predictive performance of the two models. Their results are nearly identical: the `sex`-excluded model has Acc of 0.8748, BAcc of 0.7996, Pr of 0.7783, Rc of 0.6570, and F1 of 0.7125, compared with 0.8746, 0.7990, 0.7786, 0.6557, and 0.7119, respectively, for the model that includes `sex`. Thus, excluding the explicit sensitive feature did not produce a meaningful change in the predictive performance observed on this test split.

The fairness results are shown in Table 3. The model including `sex` assigns positive predictions to 8.49% of female records and 25.59% of male records, yielding a DPD of 0.1710. After removing `sex`, the corresponding positive prediction rates are 8.85% and 25.48%, and DPD is 0.1662. The reduction in DPD is therefore small (0.0048 on the reported test split). EOD also changes only modestly, from 0.0604 to 0.0575. For the `sex`-excluded model, the TPR difference is 0.0354 and the FPR difference is 0.0575; the latter determines the reported EOD. These results show persistent group differences after removing `sex`, but they do not by themselves identify which remaining features account for those differences.

Figure 4 presents the principal performance and fairness metrics together. The performance panels reinforce the close agreement between the two models, while the DPD and EOD panels show that the fairness differences are also small relative to the observed disparity remaining in both models. Figure 5 provides the group-level view underlying the DPD values: the positive prediction rate is substantially higher for male than female records under both feature sets. Removing `sex` slightly raises the female selection rate and slightly lowers the male selection rate, but the two bars remain far apart. The figures therefore provide a visual check of the tabulated results rather than suggesting a large effect that is not present in the measurements.

Table 3: Group-specific selection rates and fairness metrics for XGBoost models trained with and without sex. Female and male test-group sizes are  $n = 5,421$  and  $n = 10,860$ , respectively. DPD and EOD are reported once per model because they summarize the two groups.

| Model       | Group  | SR     | TPR    | FPR    | DPD    | TPR diff. | FPR diff. | EOD    |
|-------------|--------|--------|--------|--------|--------|-----------|-----------|--------|
| With sex    | Female | 0.0849 | 0.6102 | 0.0207 | 0.1710 | 0.0538    | 0.0604    | 0.0604 |
| With sex    | Male   | 0.2559 | 0.6640 | 0.0811 | —      | —         | —         | —      |
| Without sex | Female | 0.0885 | 0.6271 | 0.0228 | 0.1662 | 0.0354    | 0.0575    | 0.0575 |
| Without sex | Male   | 0.2548 | 0.6625 | 0.0802 | —      | —         | —         | —      |

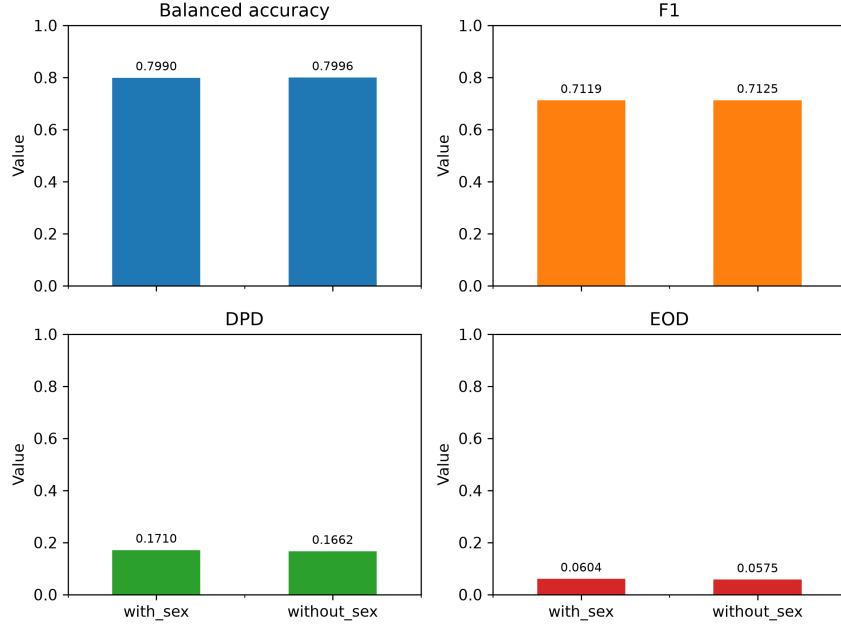


Figure 4: Predictive performance and group-fairness metrics for XGBoost models trained with and without sex. The four panels report balanced accuracy, F1, DPD, and EOD on the same held-out test split. Higher values are preferable for the predictive metrics, whereas lower values indicate smaller observed group disparity for DPD and EOD.

Taken together, Table 2, Table 3, and Figures 4–5 show that removing sex alone is insufficient to substantially change either predictive behavior or the measured group disparities in this experiment. The result is consistent with the possibility that other features retain information associated with the observed sex groups. It is not, however, evidence that any particular feature is a proxy, nor does it establish a causal explanation; those questions are addressed by the proxy-feature identification and removal analyses that follow.

## 4.2 Proxy-Feature Identification and Removal

### 4.2.1 Proxy identification

Using the sex-excluded model, we computed mean absolute TreeSHAP contribution,  $|\overline{\text{SHAP}}|$ , after aggregating one-hot contributions to the original feature. This measures model reliance. We also computed  $V_{\text{sex}}$ , defined in the evaluation metrics section, to measure feature association with the observed sex groups. Larger values indicate stronger model reliance or stronger univariate association, respectively.

These are screening diagnostics, not causal tests. We flagged a feature as a candidate proxy when it ranked in the upper half for both model reliance and sex association. Every available feature was tested; no feature was assigned to a proxy or non-proxy group in advance. Thus, an important

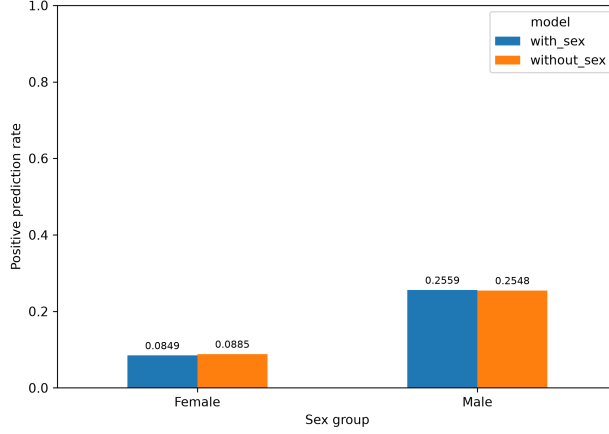


Figure 5: Positive prediction rates by observed `sex` group for XGBoost models trained with and without `sex`. Removing the feature produces only a small change in either group rate, and the male–female difference remains visible for both models.

Table 4: Diagnostics for proxy screening in the sex-excluded model. Features are ordered by decreasing mean absolute TreeSHAP contribution. A downward arrow indicates this ordering;  $V_{\text{sex}}$  is defined in Section 3.3.

| Feature        | $ \text{SHAP}  \downarrow$ | $V_{\text{sex}}$ | Proxy? |
|----------------|----------------------------|------------------|--------|
| marital-status | 1.276                      | 0.462            | Yes    |
| age            | 0.750                      | 0.100            | No     |
| occupation     | 0.612                      | 0.424            | Yes    |
| capital-gain   | 0.592                      | 0.067            | No     |
| education-num  | 0.509                      | 0.006            | No     |
| hours-per-week | 0.394                      | 0.265            | Yes    |
| relationship   | 0.292                      | 0.649            | No     |
| capital-loss   | 0.183                      | 0.042            | No     |
| workclass      | 0.145                      | 0.153            | No     |
| fnlwgt         | 0.138                      | 0.025            | No     |
| race           | 0.073                      | 0.118            | No     |
| native-country | 0.073                      | 0.057            | No     |
| education      | 0.057                      | 0.093            | No     |

predictive feature with weak sex association is not identified as a proxy, while a feature strongly associated with `sex` but weakly used by the model is not a high-priority model proxy.

Table 4 identifies `marital-status`, `occupation`, and `hours-per-week` as the features satisfying both ranking requirements. It also illustrates why both diagnostics are needed: `relationship` has the strongest sex association ( $V_{\text{sex}} = 0.649$ ) but relatively low model reliance, whereas `capital-gain` has high model reliance but weak sex association ( $V_{\text{sex}} = 0.067$ ). The result is a screening conclusion, not proof that any feature is causally a proxy.

Figure 6 provides a visual summary of model reliance. `marital-status` is the most influential feature, while `age` and `capital-gain` are also influential but are not selected because their sex-association values in Table 4 are smaller.

#### 4.2.2 Proxy removal

We next explain the intervention and its interpretation. We retrained models after removing each selected feature individually and after removing all three together. The `with-sex` and `without-sex` models provide the Step 2 and Step 3 references. Lower DPD and EOD indicate smaller observed disparities, while higher Acc, BAcc, Pr, Rc, and F1 indicate better predictive performance. Table 5 reports the held-out comparison.

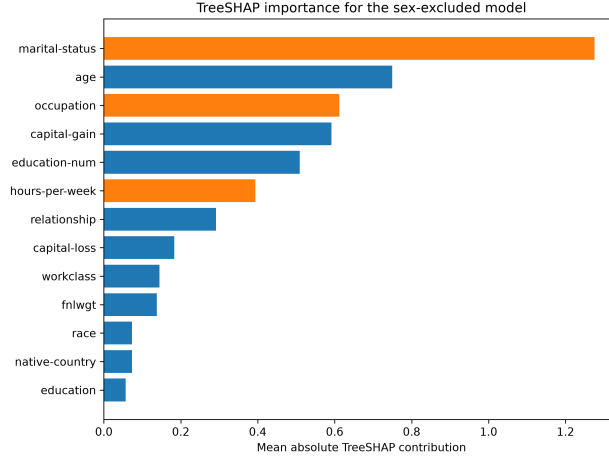


Figure 6: Mean absolute TreeSHAP contributions for the sex-excluded XGBoost model, aggregated to original features. Orange bars indicate candidate proxies under the joint screening rule; blue bars are not selected.

Table 5: Predictive and fairness outcomes after controlled proxy ablation on the fixed held-out test split ( $n = 16,281$ ).

| Model                               | Acc    | BAcc   | Pr     | Rc     | F1     | DPD    | EOD    |
|-------------------------------------|--------|--------|--------|--------|--------|--------|--------|
| With <code>sex</code>               | 0.8746 | 0.7990 | 0.7786 | 0.6557 | 0.7119 | 0.1710 | 0.0604 |
| Without <code>sex</code>            | 0.8748 | 0.7996 | 0.7783 | 0.6570 | 0.7125 | 0.1662 | 0.0575 |
| Without <code>marital-status</code> | 0.8736 | 0.7975 | 0.7763 | 0.6531 | 0.7094 | 0.1661 | 0.0580 |
| Without <code>occupation</code>     | 0.8668 | 0.7858 | 0.7631 | 0.6323 | 0.6916 | 0.1638 | 0.0607 |
| Without <code>hours-per-week</code> | 0.8730 | 0.7943 | 0.7792 | 0.6451 | 0.7058 | 0.1657 | 0.0569 |
| Without selected proxies            | 0.8690 | 0.7844 | 0.7775 | 0.6240 | 0.6923 | 0.1576 | 0.0528 |

The individual ablations in Table 5 are modest and non-uniform. Removing `marital-status` barely changes DPD or EOD; removing `hours-per-week` slightly lowers both; and removing `occupation` causes the largest individual performance loss without a clearer fairness improvement. No single feature accounts for the residual disparity on its own.

The joint removal is the clearest result. Relative to the sex-excluded model, DPD decreases from 0.1662 to 0.1576 and EOD from 0.0575 to 0.0528, while BAcc decreases from 0.7996 to 0.7844 and F1 from 0.7125 to 0.6923. The intervention therefore reduces, but does not eliminate, the disparity that remains after removing `sex`, and it does so at a measurable performance cost.

Figure 7 makes this result concrete: the sex-excluded model is close to the model using `sex`, while joint proxy removal moves toward lower DPD and lower BAcc. The analysis therefore supports a limited conclusion: removing `sex` is insufficient, and removing selected candidate proxies improves the measured disparity more than removing `sex` alone, but proxy deletion is neither cost-free nor complete. The candidates are empirical associations rather than proven causal mechanisms; interactions and the binary `sex` representation remain limitations.

### 4.3 Fairness-Constrained Retraining

## 5 Discussion

## 6 Conclusion

## References

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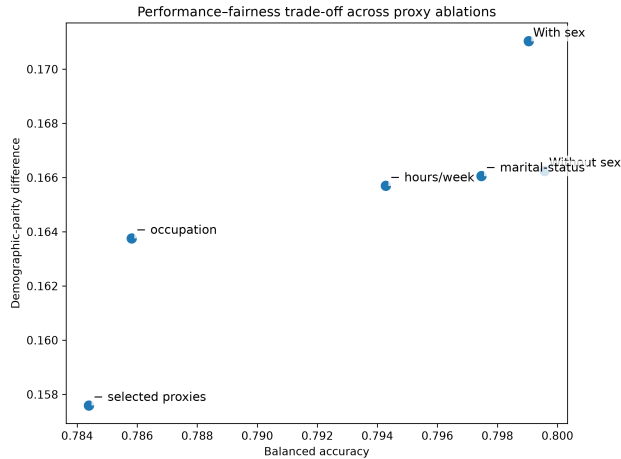


Figure 7: Balanced accuracy versus DPD for the reference and proxy-ablated models. Each point is evaluated on the same held-out test split. Lower DPD indicates smaller observed disparity; lower BACC represents a predictive performance cost.

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